

rice, meat and cake only. On de-pressing Up, the software executes manual mode sub-routine. At manual mode, total cooking time can be selected by pressing Up or Down. With each press, time increments or decrements by one second.

By holding Up or Down, time increments or decrements by 10 seconds. After setting the time, any cook mode can be selected (Micro Veg or Combi). On/off delay is according to the preset sequence, according to the type of cooking selected. Accurate delay time is achieved by enabling Timer0 of the MCU (IC2).

Unlike in conventional ovens, double protection is provided for door opening while cooking, both in software and hardware. When the door is opened, coil ground connection to magnetron relay disconnects. At the same time, software detects this condition and switches off the magnetron.

LCD1 shows all parameters. It is connected in four-wire method or 4-bit mode of operation. Corresponding software is simple and self explanatory.

At EFY Lab, Topview programmer was used for programming the IC.

Construction and testing

An actual-size PCB layout of the microwave oven is shown in Fig. 3 and its components layout in Fig. 4.

Place all components on the PCB and solder carefully. Load the program (hex file) of the software to the MCU IC. Use IC base on the PCB. Solder the tactile switches on a separate PCB and connect the same to the main PCB, as per the circuit diagram.

After assembling the circuit, enclose it in a suitable box. Fix the switches and relays on the front and back sides of the cabinet. Connect relay coils on the PCB using 2-wire cables. Connect grill heater, HV transformer, stir motor and light

using external high current-carrying wires via relay contacts.

Connect the wires to CON1, CON2, relays, CON3 and CON4 as shown in Fig. 2, except HV transformer and

heater connections. Connect two 100W lamps instead of HV transformer and grill heater. Switch on the oven, and ensure all controls are working properly.

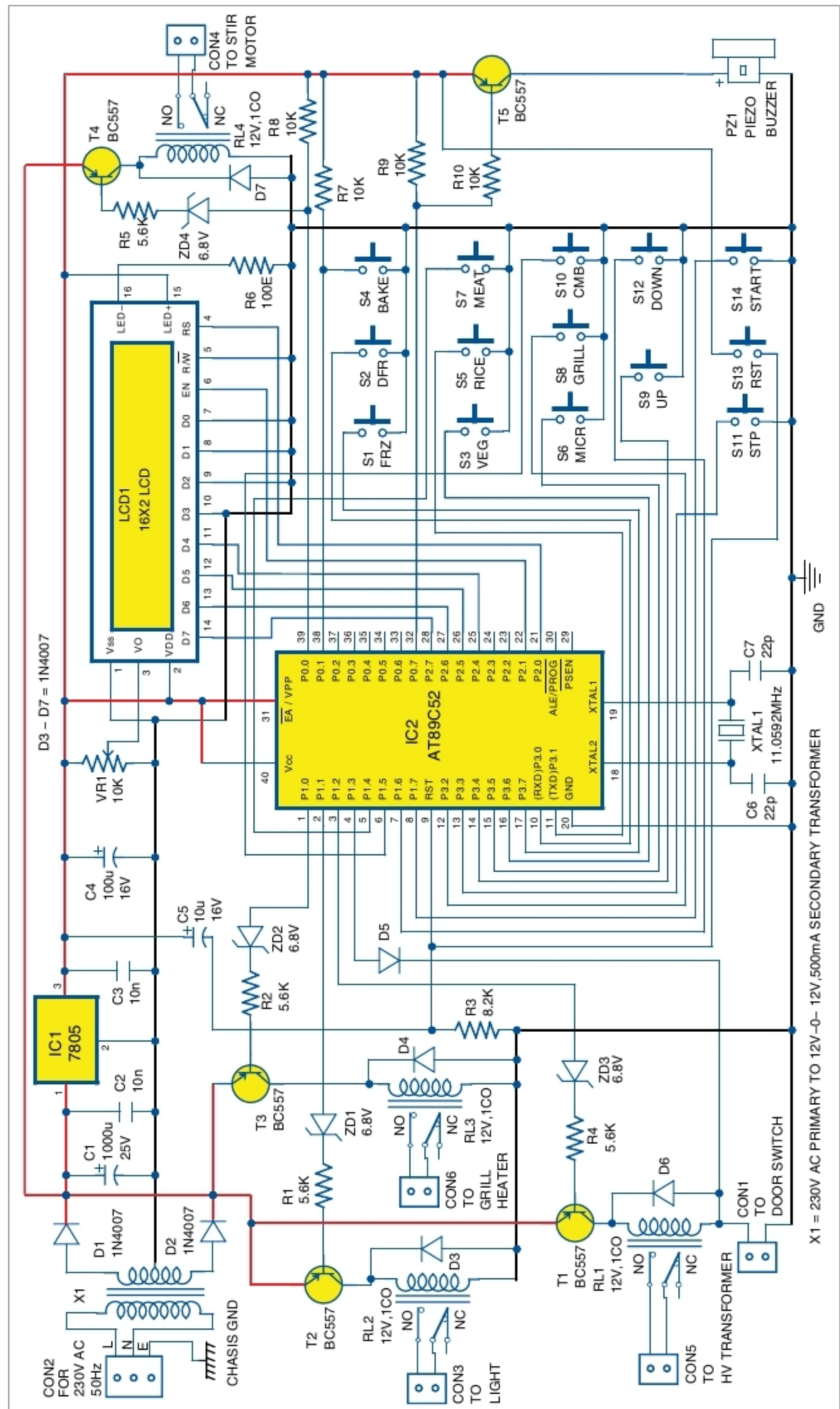


Fig. 2: Circuit diagram of microwave oven